



System Overview, Live Features & Roadmap

Orbit is an intelligent, distraction-free study operating system engineered specifically for high-stakes STEM/JEE and Computer Science scholars. It replaces static to-do lists with a closed-loop engine that automatically regulates cognitive fatigue, schedules spaced revisions, and categorizes practice errors.

⚡ Active Core Features (Ready to Use Right Now)

1. The Command Cockpit (`/planner`) **CORE**

A focused daily desk that organizes study into structured Morning, Afternoon, and Evening sessions. Features dynamic streak tracking, XP levelling, and real-time progress rings.

2. Cognitive Fatigue Shield **PROTECTION**

Monitors sleep and energy metrics. If sleep is $<6h$ or energy ≤ 2 , the engine automatically throttles high-friction problem sets and caps tactical load to prevent burnout.

3. FSRS Adaptive Spaced Repetition **MEMORY**

Calculates individualized forgetting curves. Surfaces chapters right before memory decay with an in-place active recall self-assessment (Again, Hard, Good).

4. Closed-Loop Mistake Book (`/mistakes`) **HIGH-ROI**

Classifies errors by root cause: *Conceptual, Calculation, Careless, Application, or Time Management*. Auto-schedules the parent chapter for spaced review.

5. Strict Single-Track Mastery **CURRICULUM**

Guarantees total curriculum isolation. Users are enrolled strictly in **STEM & Olympiad** (JEE Physics, Chem, Math) or **CS & AI** (DSA, AI/ML, OS/DBMS).

6. Zero-Latency Local-First Sync **ARCHITECTURE**

Backed by client IndexedDB (Dexie) for 0ms offline interaction, automatically synced with Supabase PostgreSQL cloud storage with Row-Level Security.

🚧 What Can Be Implemented Later (Product Roadmap)

Orbit is architected with clear abstraction boundaries to effortlessly accommodate these planned advanced modules:

AI / ML

Deep Reasoning Mentor

Wired LLM inference (Claude / DeepSeek API) to diagnose root conceptual misconceptions directly from logged mistake descriptions.

ANALYTICS

Focus Duration Insights

Automated telemetry utilizing the existing `study\_session` schema to detect focus-fade patterns (e.g. "Cognitive efficiency dips after 78 min").

VISION

OCR Mistake Ingestion

Snap a photo of handwritten test scratchwork. Auto-converts math into LaTeX equations and auto-detects calculation flaws.

SOCIAL

Peer Focus Circles

Live mutual study rooms utilizing the `orbit\_partner\_name` schema for real-time Pomodoro accountability and silent co-working.

SYNC

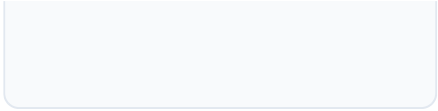
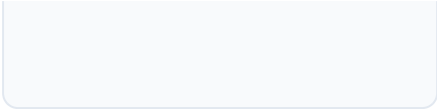
Anki Deck Export

One-click export of unresolved mistakes and key review formulas into standard Anki `.apkg` packages for mobile card drilling.

SIMULATOR

Live Mock Test Mode

Full-screen timed test interface simulating the exact NTA JEE / GATE



testing terminal with post-exam
mistake auto-logging.



How to Use `/planner` (The Command Desk)

The Planner is your primary daily cockpit. You spend 90% of your active study day on this screen. It eliminates decision fatigue by showing you exactly what to study right now.

1. Managing Daily Tasks

- 1 Add a Mission:** Click "+ Add Task". Enter a title (e.g. "Solve 25 Rotational Dynamics MCQs"), choose a subject/chapter, assign a time slot, and set estimated minutes.
- 2 Check Off Completed Work:** Tap the circle next to any task. Completed tasks trigger positive reinforcement, increment XP, and feed the daily completion ring.
- 3 Reschedule or Move:** Swipe or click task options to move incomplete tasks to tomorrow without breaking your streak.

2. The Daily Orbit Ring

The glowing circular ring at the top displays your percentage completion for the current calendar date. Hitting 100% awards bonus streak protection and unlocks daily mastery levels.

3. Deep Work Focus Timer

Click the **Focus Timer** on any task to enter an immersive Pomodoro focus session. It tracks uninterrupted study minutes and records focus sessions to your local database.

The Fatigue Shield in Action

Click "**Calibrate Plan**" or log your morning check-in:

- **Sleep < 6.0 Hours:** The engine flags cognitive depletion, caps your daily target to 4 hours max, and hides 3-hour hard problem sets.
- **Low Energy ($\leq 2/5$):** Shifts focus to active recall and summary reading rather than new conceptual heavy-lifting.

4. The Spaced Revision Chamber

When chapters are due for retention review according to the FSRS forgetting curve, they appear in the **Active Recall Chamber** on your desk:

Self-Assessment Grade	What It Means	Algorithm Impact
Again	Forgot key concepts or formulas	Resets review interval to 1 day; flags for urgent review.
Hard	Recalled with heavy mental strain	Expands interval cautiously (e.g., 2-3 days).
Good	Fast, confident recall	Expands interval exponentially (e.g., 7-14 days).

You can complete revisions right on your planner desk in under 3 minutes.



How to Use `/journey` (Syllabus Mastery Map)

The Journey tab gives you a high-altitude visual radar of your entire syllabus. It tracks your progression from complete novice to full exam-level mastery for every chapter.

1. Single-Track Syllabus View

Your Journey tab exclusively displays the curriculum of your enrolled track:

- 1 **STEM Track:** Physics (Mechanics, Thermo, Electrodynamics, Optics), Chemistry (Organic, Physical, Inorganic), Mathematics (Calculus, Vectors, Algebra).
- 2 **CS Track:** Data Structures & Algorithms, Machine Learning & Deep Learning, Database Management Systems, Computer Networks & OS.

2. Chapter Mastery Lifecycles

Every chapter moves through 5 distinct academic stages:

Status	Meaning & Action Required
NOT STARTED	Untouched material. Ready for initial lectures.
LEARNING	Currently attending classes or reading theory.
PRACTICING	Solving PYQs and standard problem sheets.
REVISION DUE	Memory fading! FSRS algorithm has triggered a review.
MASTERED	High confidence (>85%) with zero unresolved errors.

3. Confidence Sliders & Quick Update

Click on any subject card to expand its chapters. For each chapter:

- 1 Adjust the **Confidence Slider (0% - 100%)** based on how comfortable you feel solving unassisted exam problems.
- 2 Change the status badge with 1 tap.
- 3 Observe the real-time subject mastery percentage bar update automatically.

4. Chapter Detail Screen (`/journey/[id]`)

Clicking directly on any chapter title opens its comprehensive **Academic Dossier**:

- **Mistake History:** View all logged errors specific to this chapter.
- **FSRS Stability Metrics:** View ease factors, review intervals, and next scheduled review date.
- **Chapter Study Notes:** Store key formula cheat-sheets, edge cases, and personal insights.

5. Adding Custom Topics

Need to add a specific college elective or coaching-specific submodule? Click "+ **Add Subject**" or "+ **Add Chapter**" to expand your syllabus anytime.



How to Use `/mistakes` (The Mistake Vault)

Logging practice and mock-test mistakes is proven to be 3x more effective for score improvement than passively re-reading notes. Orbit turns your mistakes into an active retention asset.

1. How to Log an Error (The 30-Second Habit)

Whenever you miss a question in a practice session or mock exam:

- 1 Navigate to `/mistakes` and tap "+ Log Mistake".
- 2 Select the **Subject and Chapter** the question belonged to.
- 3 Type a concise description of the question and the blunder.
- 4 Tag the **Difficulty** (Easy, Medium, Hard).
- 5 Select the precise **Root-Cause Taxonomy**.

Closed-Loop Automation

The moment you submit a mistake, Orbit automatically enrolls that chapter into the Spaced Revision queue. You will never have to manually remember to revisit your weak areas.

2. The 5-Point Root Cause Taxonomy

Orbit forces you to diagnose *why* you missed the question so you can cure the root symptom:

Taxonomy Tag	Prescription & Remedy
Conceptual	Fundamental gap in theory. Re-read core theorem or re-watch lecture derivation.
Calculation	Arithmetic or algebraic sign slip. Practice scratchwork neatness & mental math checks.
Careless	Misread question prompt (e.g. "which is NOT true"). Highlight keywords during test.
Application	Knew formula, but failed to recognize multi-concept connection in unfamiliar context.
Time Pressure	Panicked under clock countdown. Drill timed 15-minute speed sprints.

3. Resolving Mistakes

Once you re-attempt a question a week later and solve it cleanly unassisted, click "**Resolve**". It moves out of your active backlog while permanently preserving your error telemetry.

How to Use `/week` & `/profile`

These final tabs zoom out to manage your weekly tactical rhythm, upcoming exam deadlines, health baselines, and local-first data backups.

1. Tab 4: `/week` (Tactical Load Radar)

Provides a bird's-eye view of your next 7 days:

- 1 **Weekly Load Distribution:** Visualizes scheduled study hours per day to prevent front-loading Monday and burning out by Thursday.
- 2 **Exam Horizon:** Displays countdown clocks to your registered milestones (e.g., "JEE Main Session 1 — 94 days away" or "Semester Midterm").
- 3 **Syllabus Coverage Blueprint:** Displays what percentage of the test syllabus you have reached *Practicing* or *Mastered* status on.
- 4 **Add Test / Mock:** Click "+ Add Exam" to schedule coaching tests and map them to relevant syllabus chapters.

2. The Ideal 3-Minute Daily Routine

Morning (1 Min): Open Orbit on your phone/laptop. Review your daily 3 missions on `/planner`.

Midday: Start the Focus Timer during study blocks. Check off finished tasks.

Evening (2 Mins): Log errors from practice sets into `/mistakes`. Spend 2 minutes in the Spaced Revision Chamber. Sleep well!

3. Tab 5: `/profile` (Health & OS Settings)

- 1 **Academic Track Identity:** View your active enrollment (STEM vs CS). Orbit strictly isolates curricula based on this preference.
- 2 **Accountability Circle:** Set your study partner's name (`partnerName`) for shared focus momentum.
- 3 **Streak & XP Status:** View your cumulative study streak, experience points, and rank tier.

4. Data Backups & Offline Portability

100% Data Sovereignty

Orbit stores everything on your device first via IndexedDB with automatic cloud sync to Supabase Postgres. You always own your data:

- **Export Backup:** Downloads an encrypted `.json` file containing all tasks, mistakes, and revisions.
- **Import Backup:** Instantly restores your entire academic state on any new laptop or browser with 1 click.

5. Mobile PWA Installation

To use Orbit like a native app on your phone:

- 1 **iPhone (Safari):** Tap *Share* → "Add to Home Screen".
- 2 **Android (Chrome):** Tap *Install App* from the prompt.
- 3 Opens full-screen with 0ms loading lag and works completely offline!